



PURWADHIKA JCDS 2904

MACHINE LEARNING FOR CHURN PREDICTION: MAXIMIZING CUSTOMER RETENTION IN E- COMMERCE

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Executive Summary

- In the e-commerce industry, acquiring a new customer can cost **5–25 times** more than retaining an existing one. Customer churn, **when a user stops making purchases within a certain period** can reduce revenue, lower customer lifetime value, and waste marketing budgets.
- In this dataset, **churn is labeled as 1 (customer churned) and 0 (customer not churn)**. Around 16.8% of customers churn, meaning roughly 1 in 6 customers leave the platform.
- **Top churn drivers** include low cashback, complaints, low order activity, long delivery distances, and suspiciously high satisfaction scores.
- **We recommend** targeted cashback for new or low-tenure users, faster complaint resolution, delivery improvements for remote areas, and personalized outreach for inactive customers.
- Using machine learning, we built a predictive model with **91% accuracy** in identifying high-risk customers, who are likely to churn.

The marketing team hopefully can run smarter, data-driven retention campaigns, ensuring resources **focus on customers who matter most** and ultimately boosting loyalty.





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INTRODUCTION

Background Context

- E-commerce has transformed how customers shop, offering **convenience**, **competitive pricing**, and **endless product variety**.
- However, this also means customers can **easily switch to competitors** with just a few clicks.
- Retaining existing customers is therefore critical, especially since acquiring a new customer **can cost 5 to 25 times more than keeping one**.

- Given the cost and impact of churn, **two critical questions emerge;**





Problem Statement

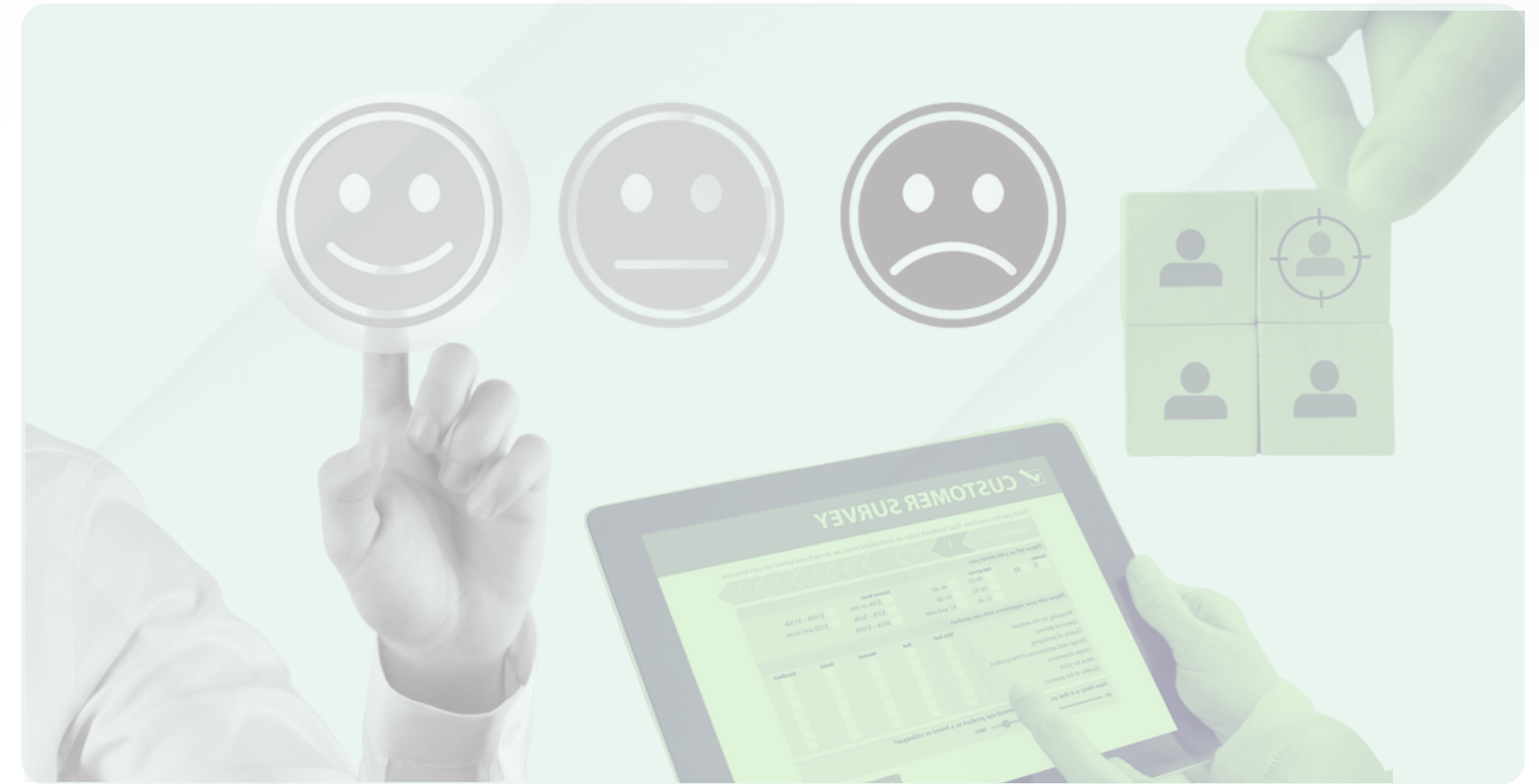


- *How* can we identify customers who are **likely** to *churn*?
- *What* are the key factors **influencing** customers to *churn*?



What are the Goals?

- To **identify customers** who are **likely to churn** by developing an accurate predictive model.



- To **determine the key factors** influencing customers who are likely to churn



Analytical Approach & Metric Evaluation

- In this project, customer churn refers to when a customer stops purchasing from our platform within a given period. In this dataset, churn is labeled as 1 for customers who have churned and 0 for customers who have not churned.

Churn matters because acquiring a new customer can cost 5 to 25 times more than retaining an existing one. Without effective churn prevention, businesses risk losing not only revenue but also long-term customer value.

To predict churn, we experimented with multiple classification models and evaluated them using the F2 Score.

	Predicted: Not Churn (0)	Predicted: Churn (1)
Actual: Not Churn (0)	True Negative (TN) Model predicts the customer will stay, and they do	False Positive (FP) Model predicts churn, but the customer actually stays
Actual: Churn (1)	False Negative (FN) Model predicts the customer will stay, but they actually churn	True Positive (TP) Model correctly predicts churn

In this project, false negative are considered the most dangerous. Given these consequences, the model should **focus on minimizing false negative**. In other words, it's more important to correctly identify customers who are likely to churn, even at the risk of a few false positive. Therefore, the primary metric for model evaluation is the F2-Score, which gives more weight to recall than precision.

STAKEHOLDERS

CUSTOMER MARKETING TEAM

- The Customer Marketing Team are directly responsible for developing and implementing strategic marketing plans to drive customer engagement and loyalty.





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EXPLORATORY DATA ANALYSIS

Data *Understanding*



Customer Demographics

CustomerID	Unique identifier, carries no numerical or ordinal meaning
City Tier	City classification with inherent ranking (1,2,3)
Warehouse To Home	Distance, measurable and can include decimals
Gender	Categories (Male, Female), not ordered
Number Of Device Registered	Count of devices, whole numbers only
Marital Status	Categories (Single, Married, Divorced.), no ranking
Number Of Address	Count of addresses, integers only

Customer Purchase History

Order Amount Hike From last Year	Percentage change, measurable and fractional
Coupon Used	Count of coupons used, integers only
Order Count	Number of orders placed, count data
Day Since Last Order	Number of days since last order, integers only
Cashback Amount	Cashback, decimal values possible

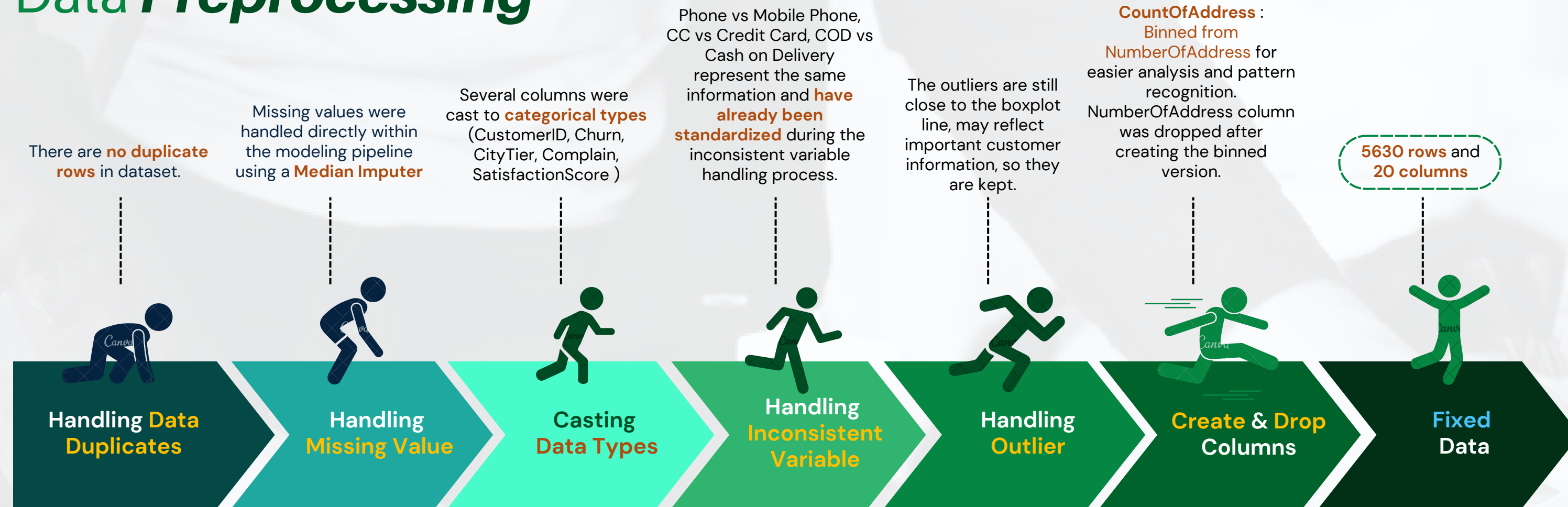




Customer Behaviour/ App usage	
<i>Preferred Login Device</i>	Device types like Mobile/Desktop, no inherent order
<i>Preferred Payment Mode</i>	Categories like Card, COD, etc., there are no ranking
<i>Prefered Order Cat</i>	Product category preference, no meaningful order
<i>Hour Spend On App</i>	Time in hours, can be fractional
<i>Tenure</i>	Duration measured in time, can be fractional
Customer Feedback/ Satisfaction	
<i>Satisfaction Score</i>	Rating scale, ordered values (1 to 5)
<i>Complain</i>	Binary indicator (1 = yes, 0 = no), no ordinal meaning
Label/Target	
<i>Churn</i>	Binary category (1 = churned, 0 = not churned)



Data *Preprocessing*



Data *Analysis*



seaborn

matplotlib



+ a b l e a u

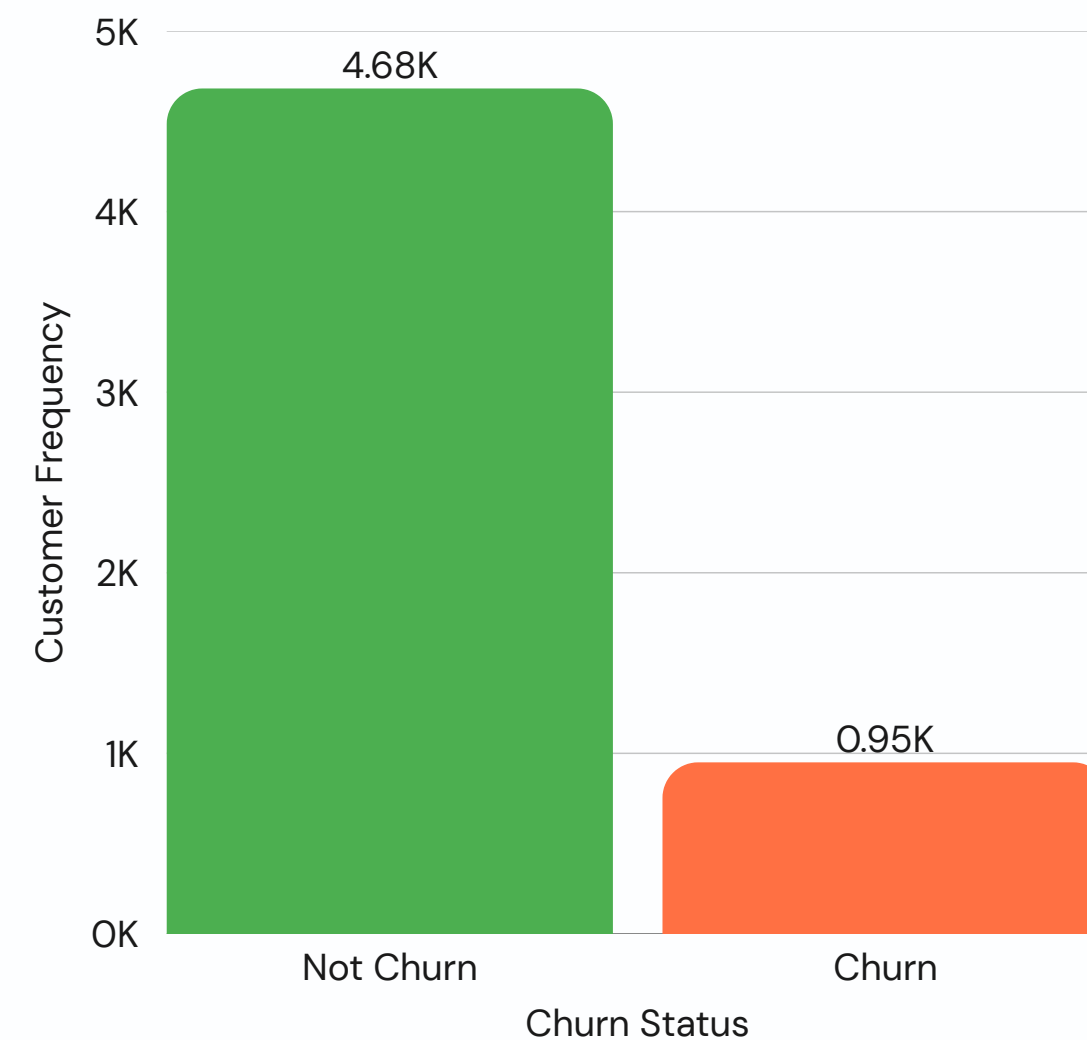
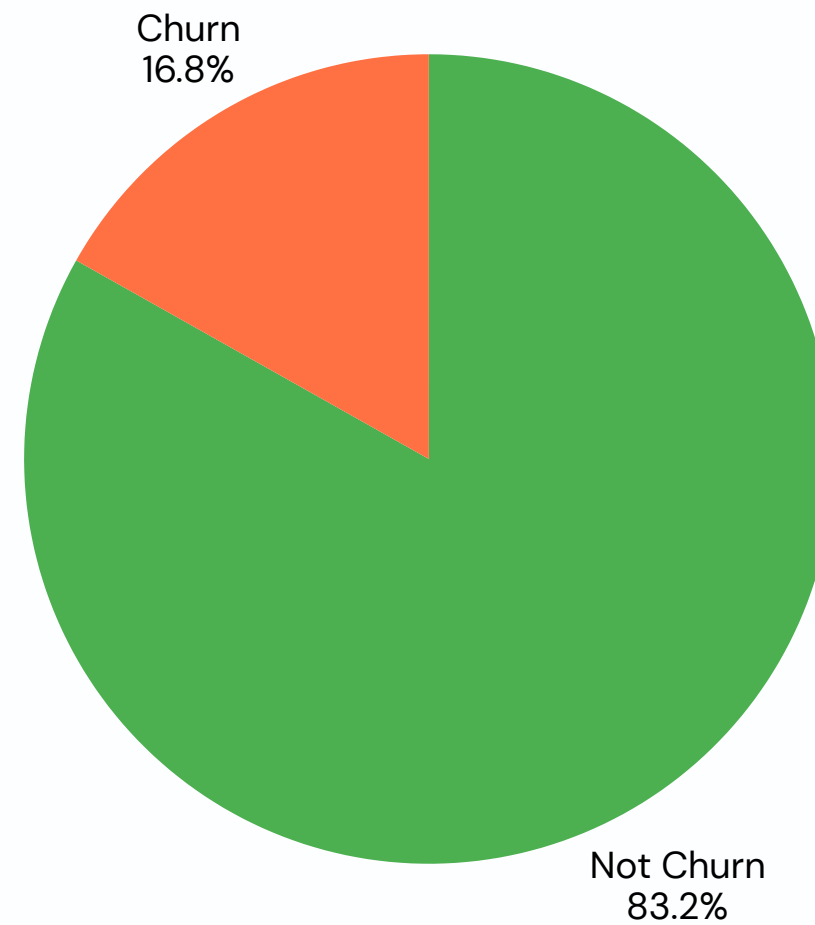


Exploratory Data *Analysis*



Overall *Churn*

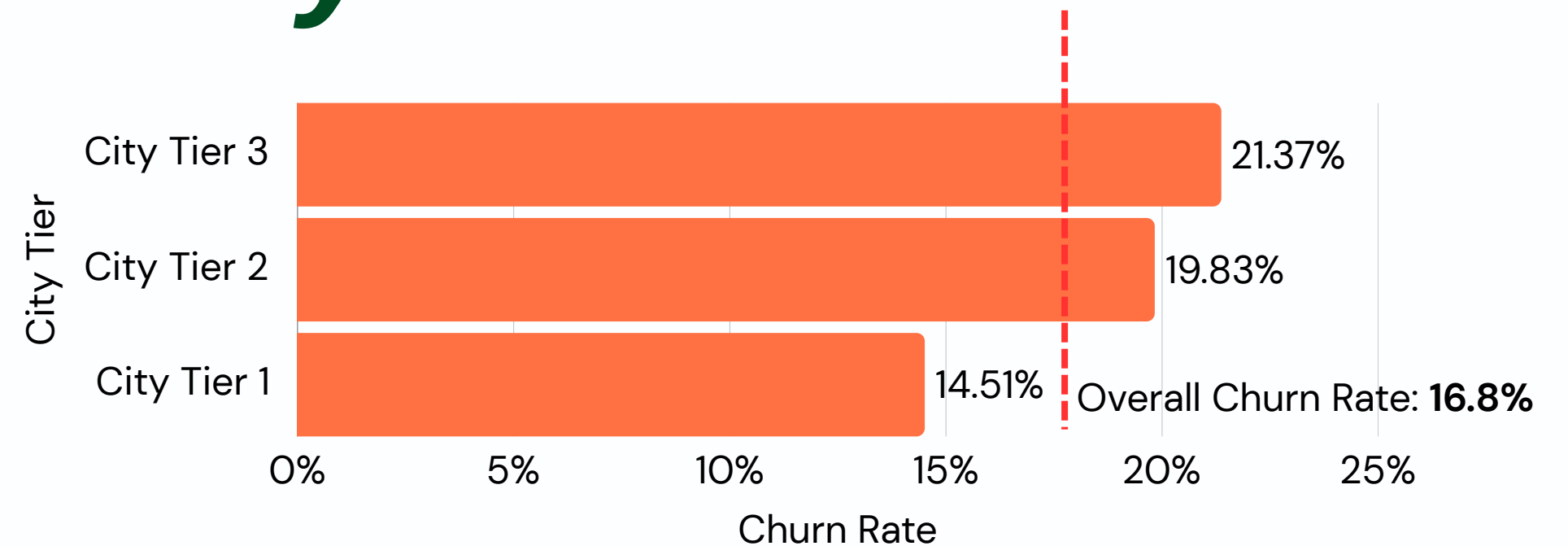
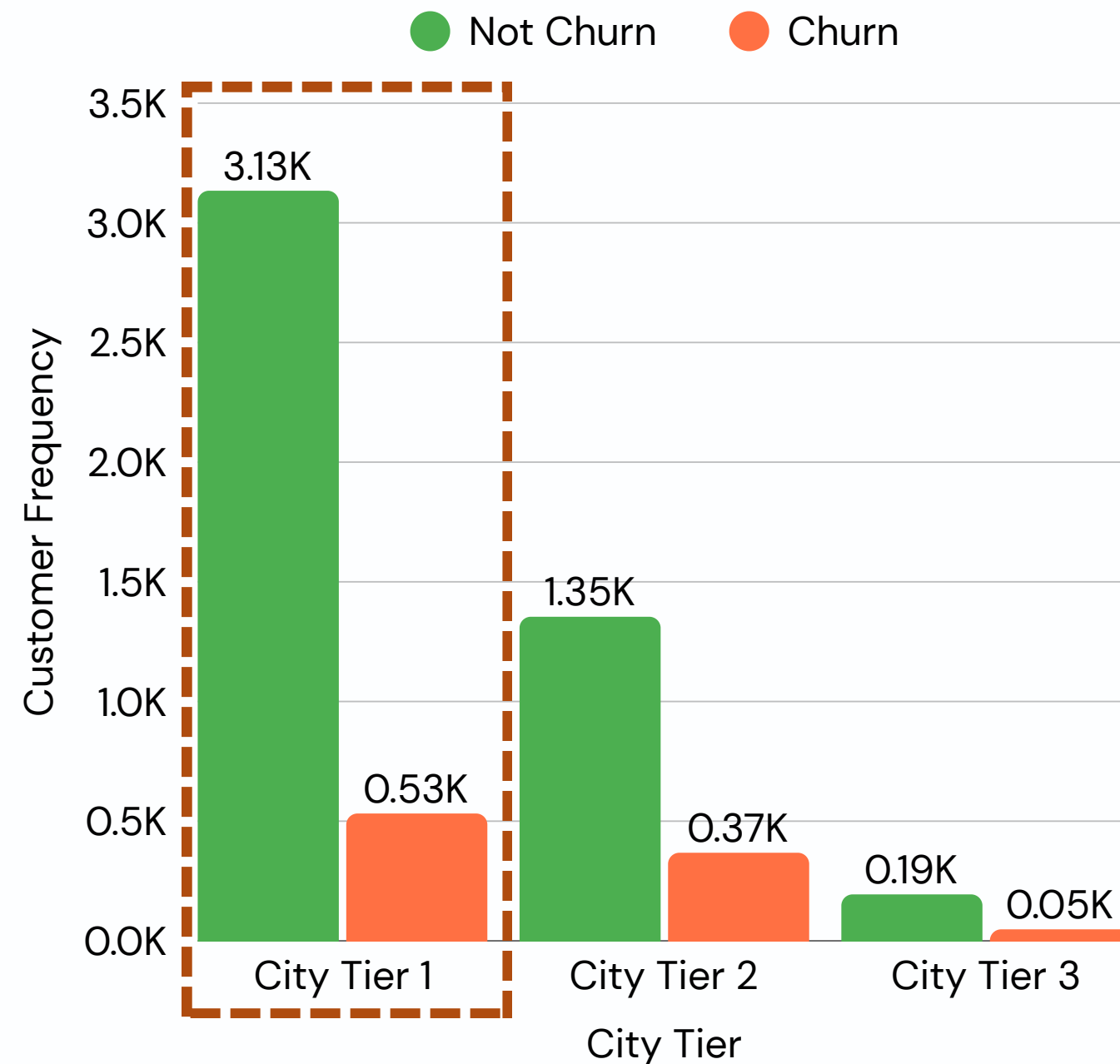
The charts show that **16.8%** of customers **have churned**, while **83.2%** **not churn**. Although the churn rate is relatively low, it still represents a significant number of customers.





Exploratory Data Analysis

Churn Rate by City Tier



Insights:

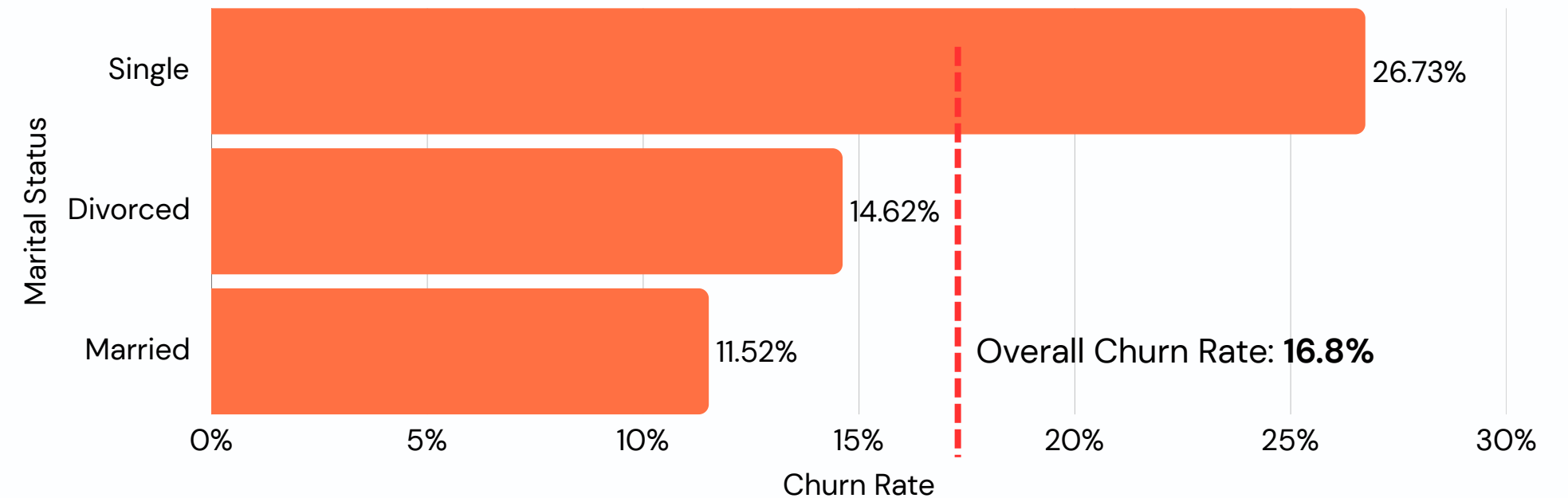
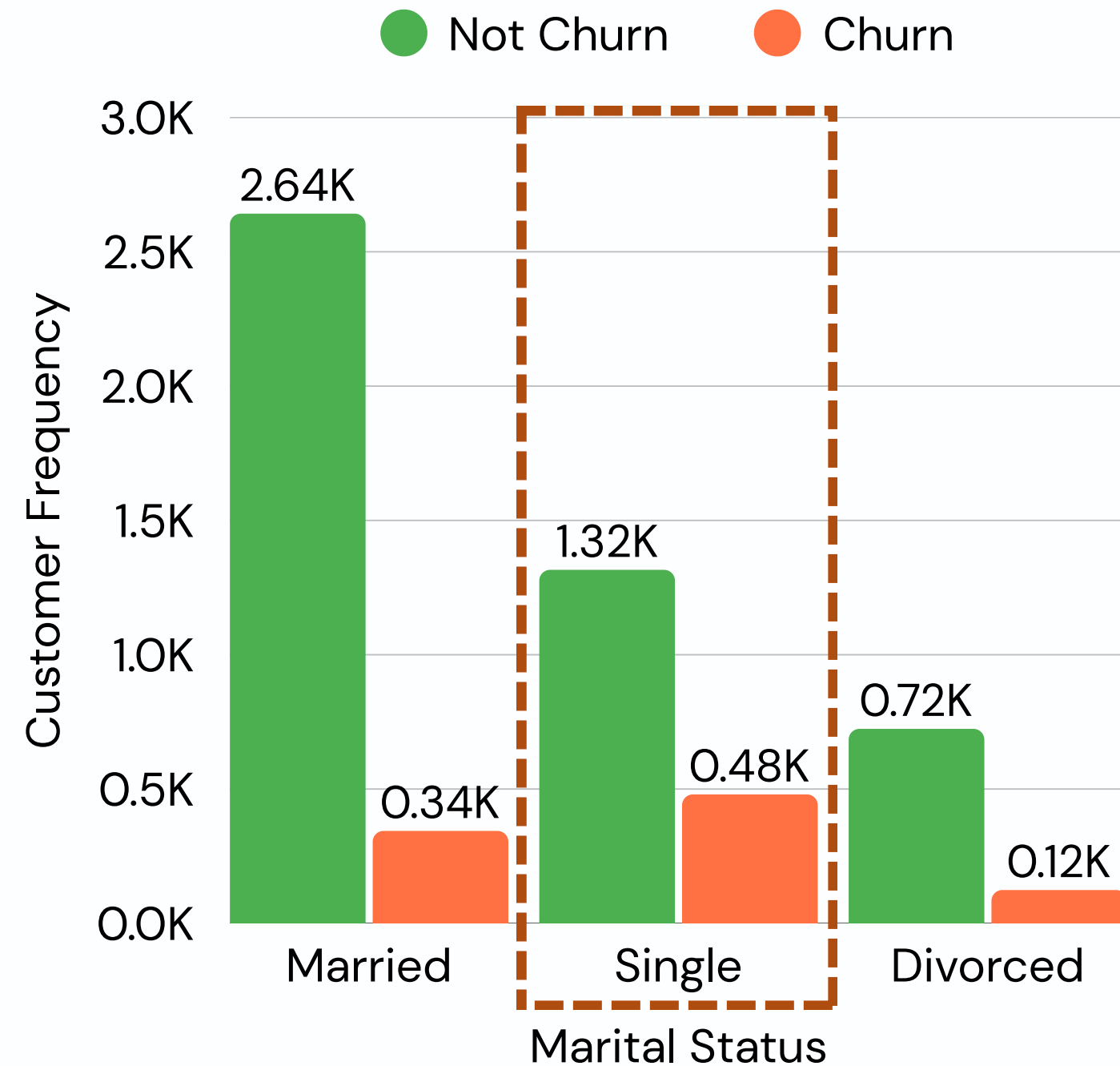
- City Tier 3 customers have the highest churn rate (21.37%), followed by Tier 2 (19.83%) — both above the overall average of 16.8%.
- City Tier 1 customers show the lowest churn rate (14.51%), indicating stronger retention.

Customers in smaller or less developed cities are more likely to churn. Focusing retention programs on City Tier 2 and 3 customers could reduce the overall churn rate.



Exploratory Data Analysis

Churn Rate by Marital Status



Insights:

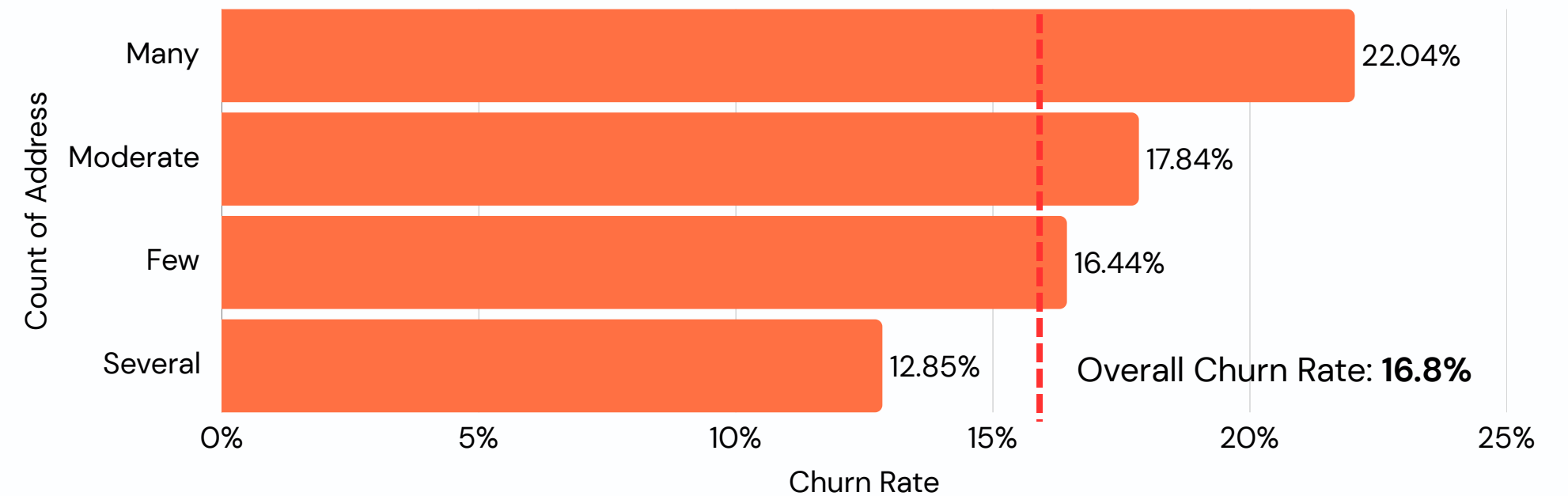
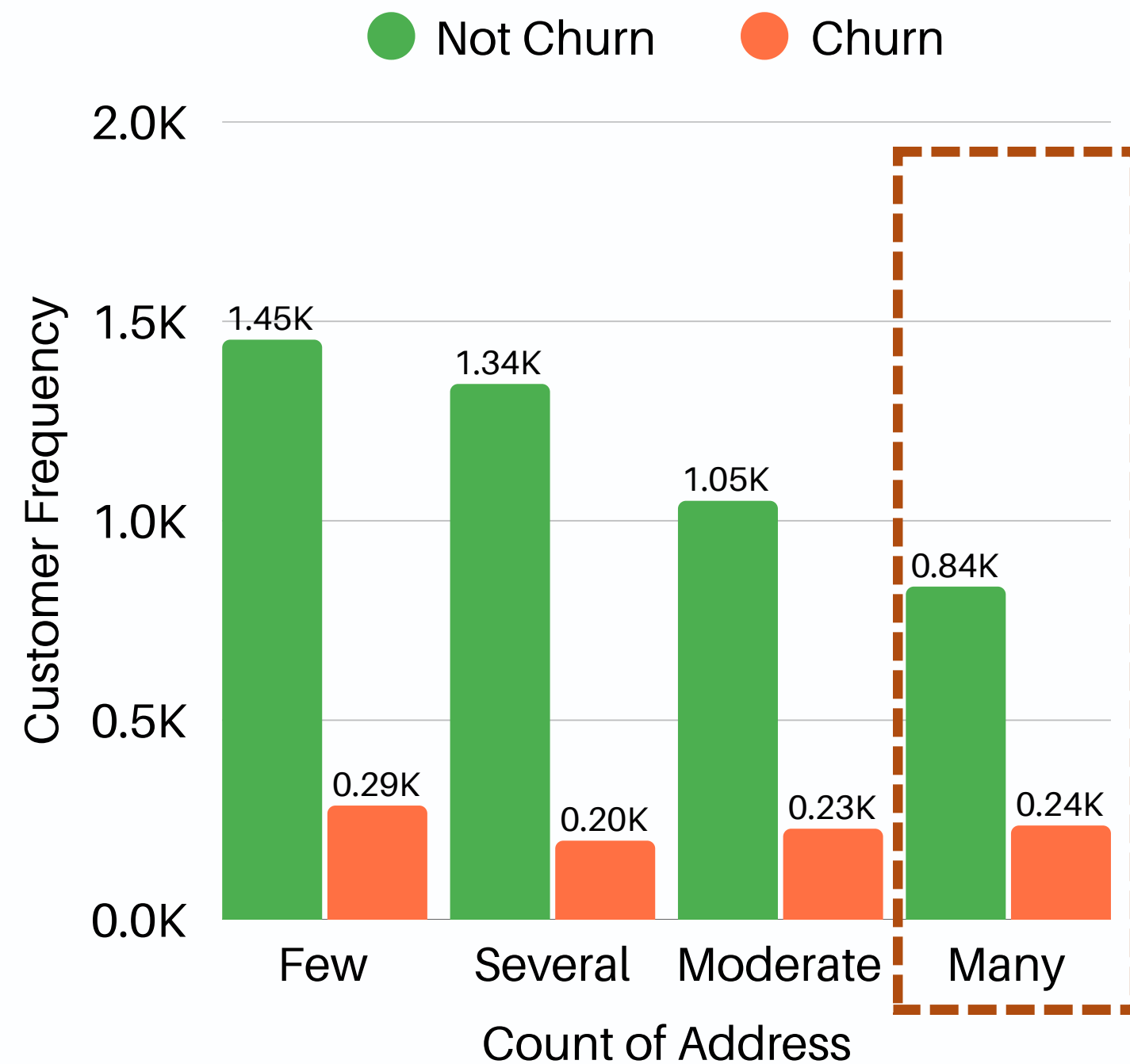
- Single customers have the highest churn rate (26.73%), far above the overall average (16.8%).
- Married customers have the lowest churn rate (11.52%), showing stronger loyalty.

Singles are at the highest risk of churning. Retention strategies should prioritize single customers to make the biggest impact on churn reduction.



Exploratory Data Analysis

Churn Rate by Count of Address



Insights:

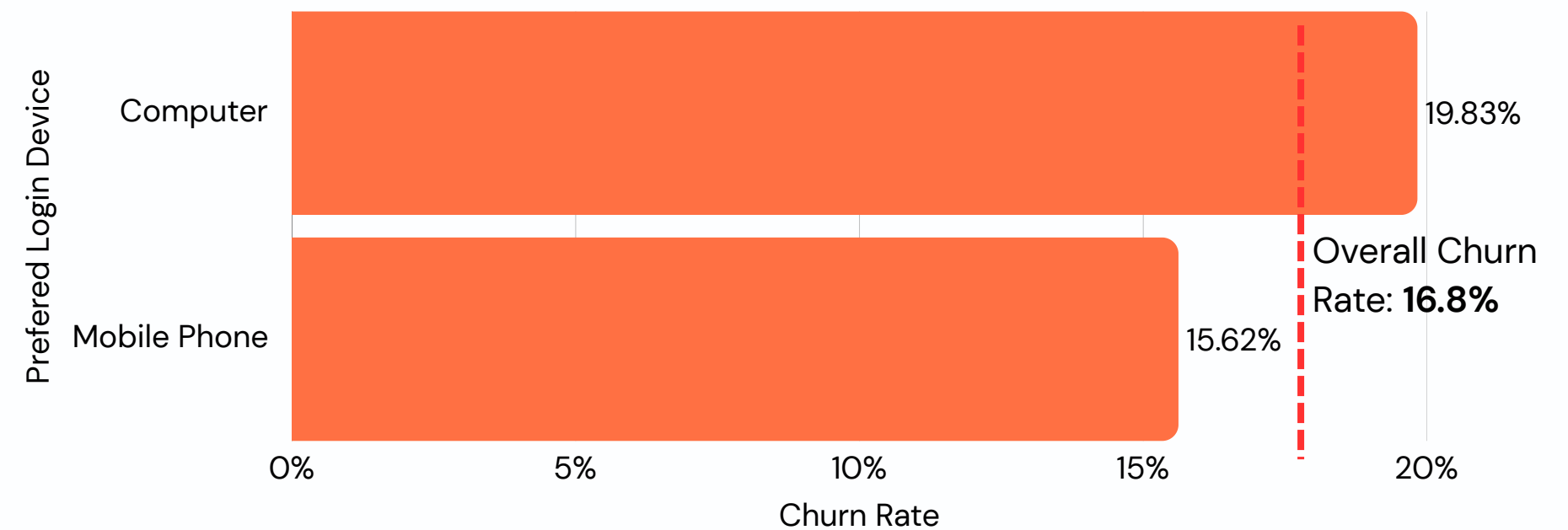
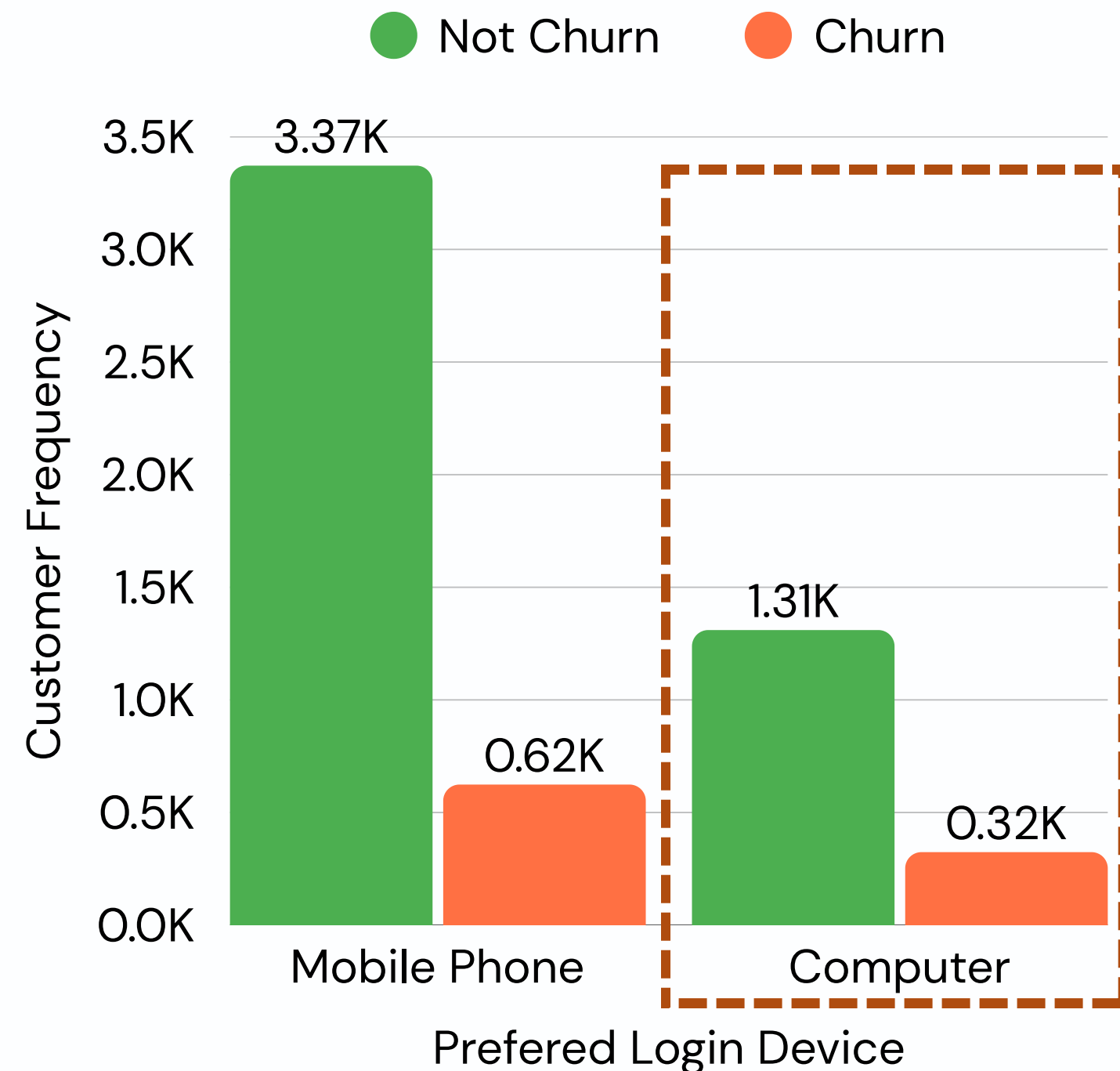
- 22.0% of customers with many addresses churn — the highest among all groups.
- Customers with several addresses have the lowest churn (12.85%).

Frequent address changes are linked to higher churn risk. Retention strategies should be focused on customers with multiple addresses, as they are at the highest risk of churn



Exploratory Data Analysis

Churn Rate by Preferred Login Device



Insights:

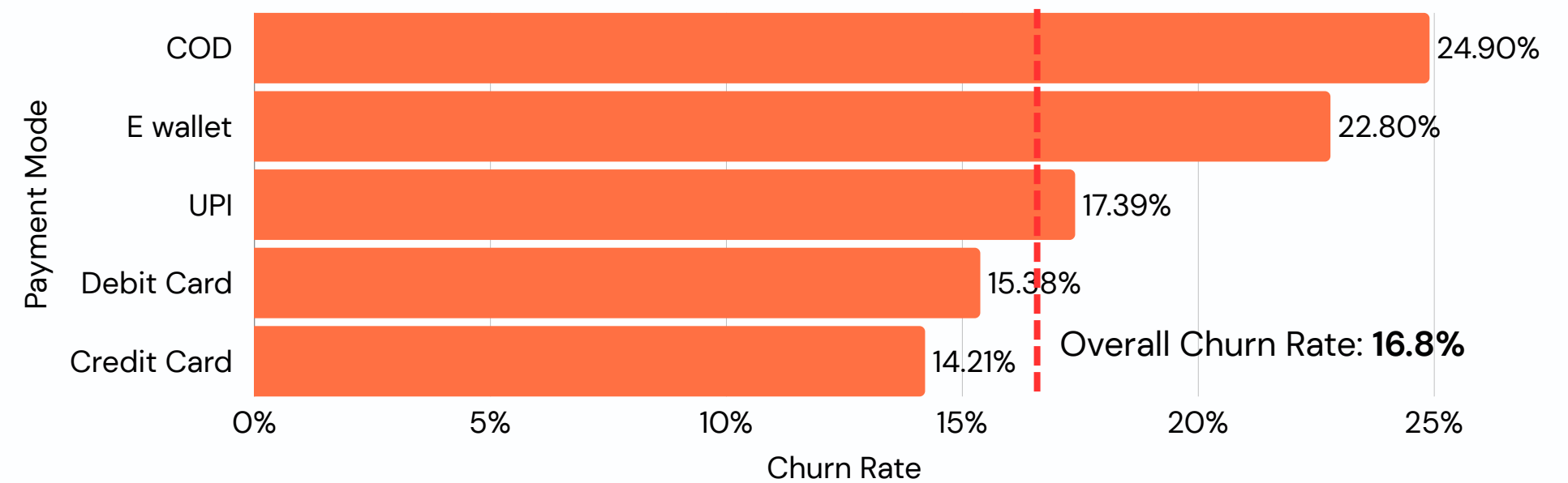
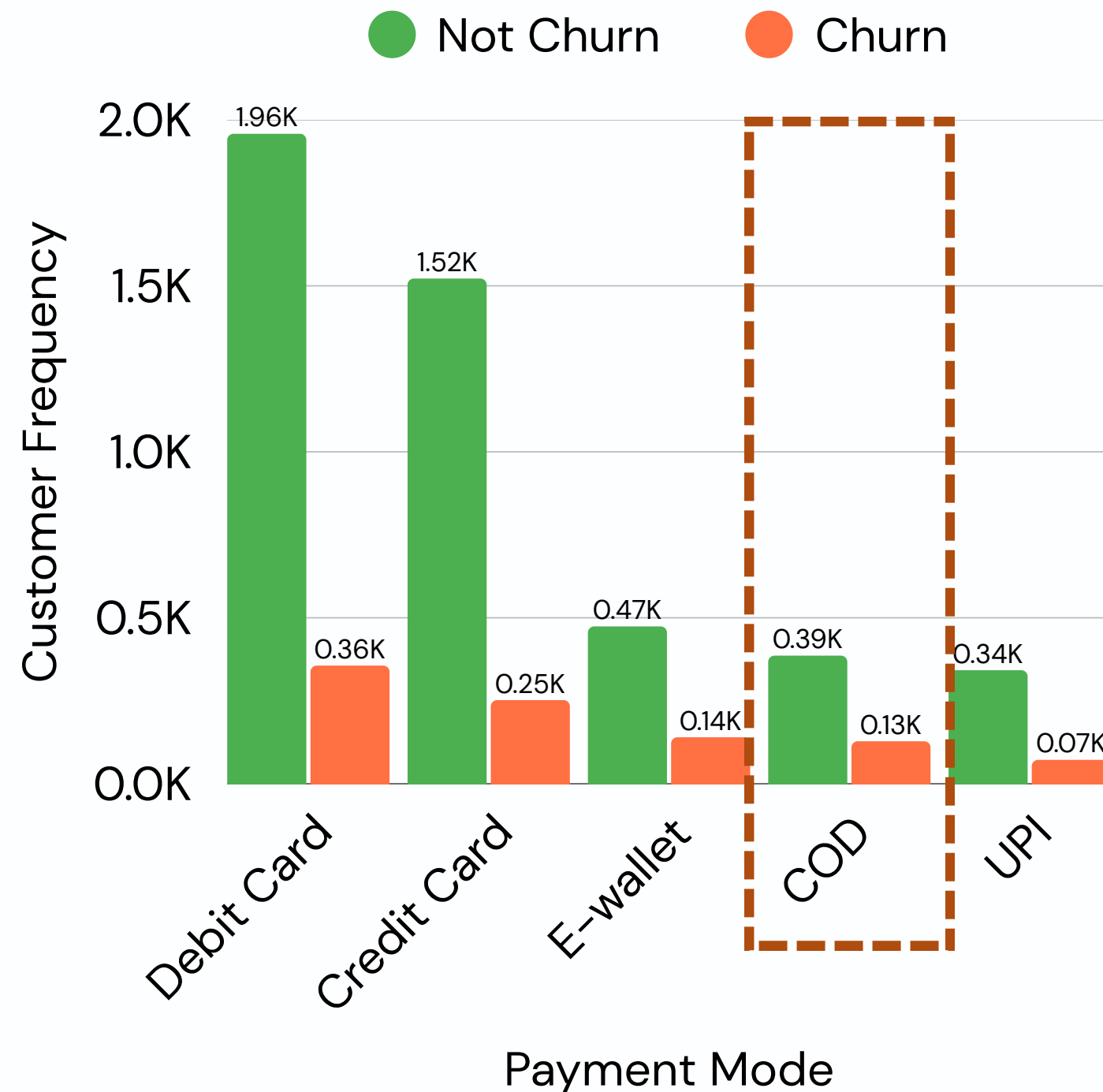
- Customers who log in via computer have a higher churn rate (18.83%) than mobile users (15.62%).
- Possible causes include less optimized UI, slower navigation, or fewer features compared to mobile.

Evaluate computer app user experience and optimize features to reduce churn among computer users.



Exploratory Data Analysis

Churn Rate by Preferred Payment Mode



Insights:

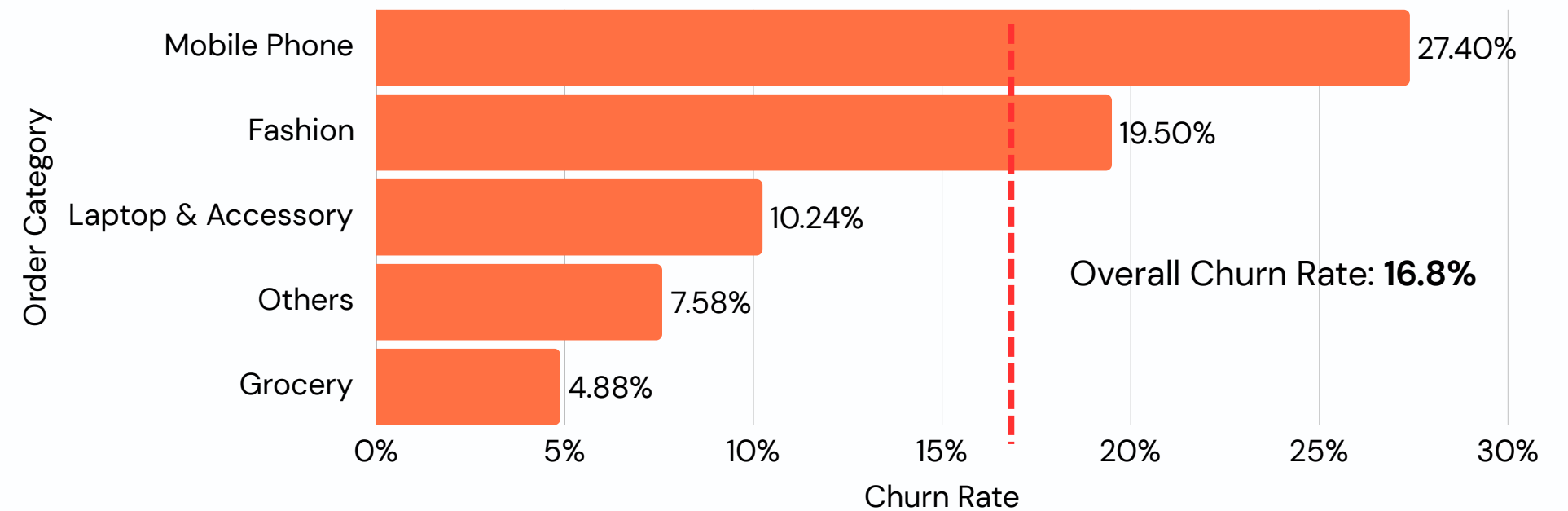
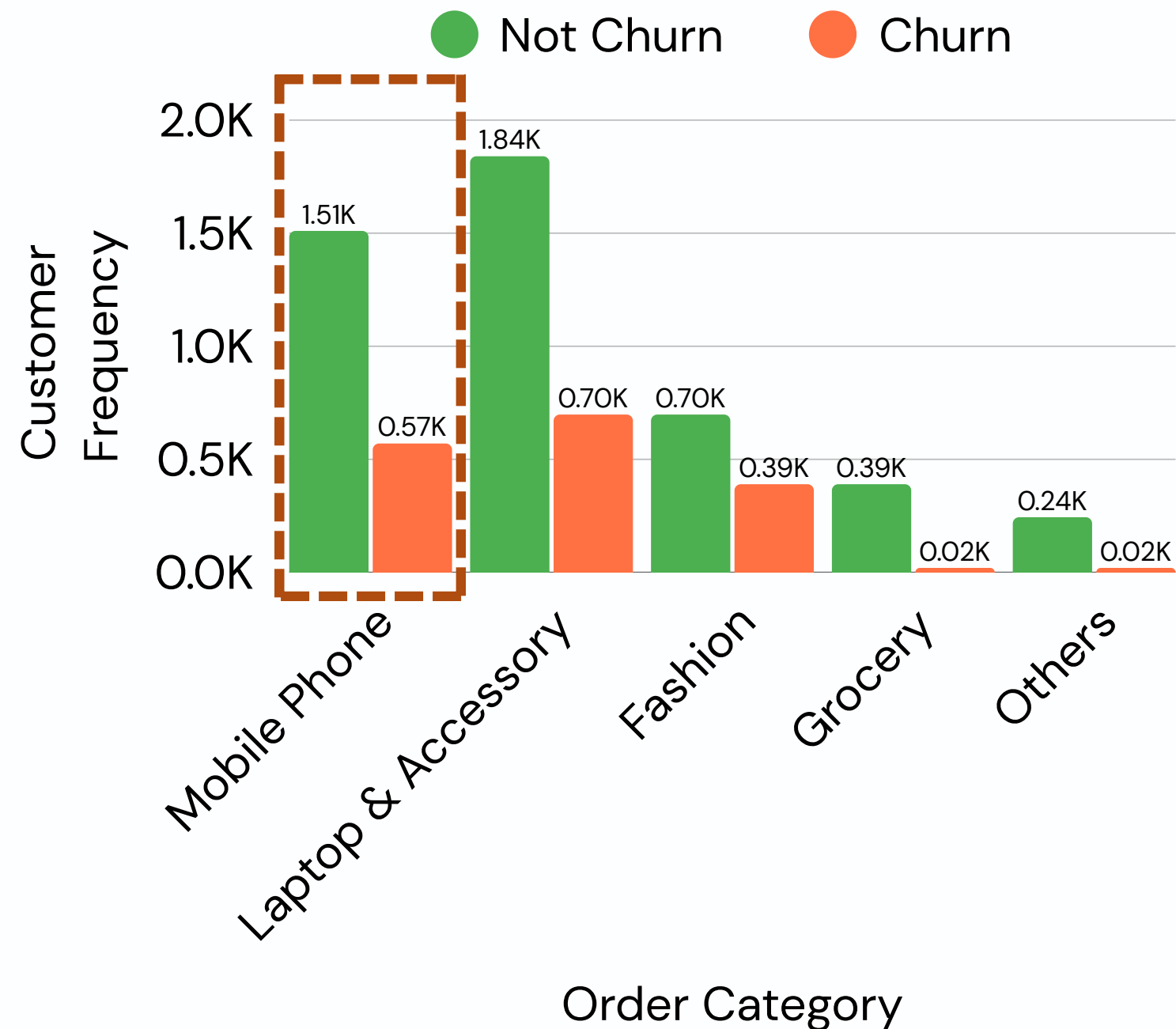
- Customers using Cash on Delivery (COD) have the highest churn rate at 24.9%, followed by E-wallet users at 22.2%
- COD churn rate is 1.48× higher than the overall churn rate (16.8%).
- 1.75 times higher than customers paying by credit card (lowest churn rate).

Evaluate COD and E-wallet payment process and provide incentives to shift COD and E-wallet users to more secure and convenient payment modes such as credit/debit cards or UPI.



Exploratory Data Analysis

Churn Rate by Preferred Order Category



Insights:

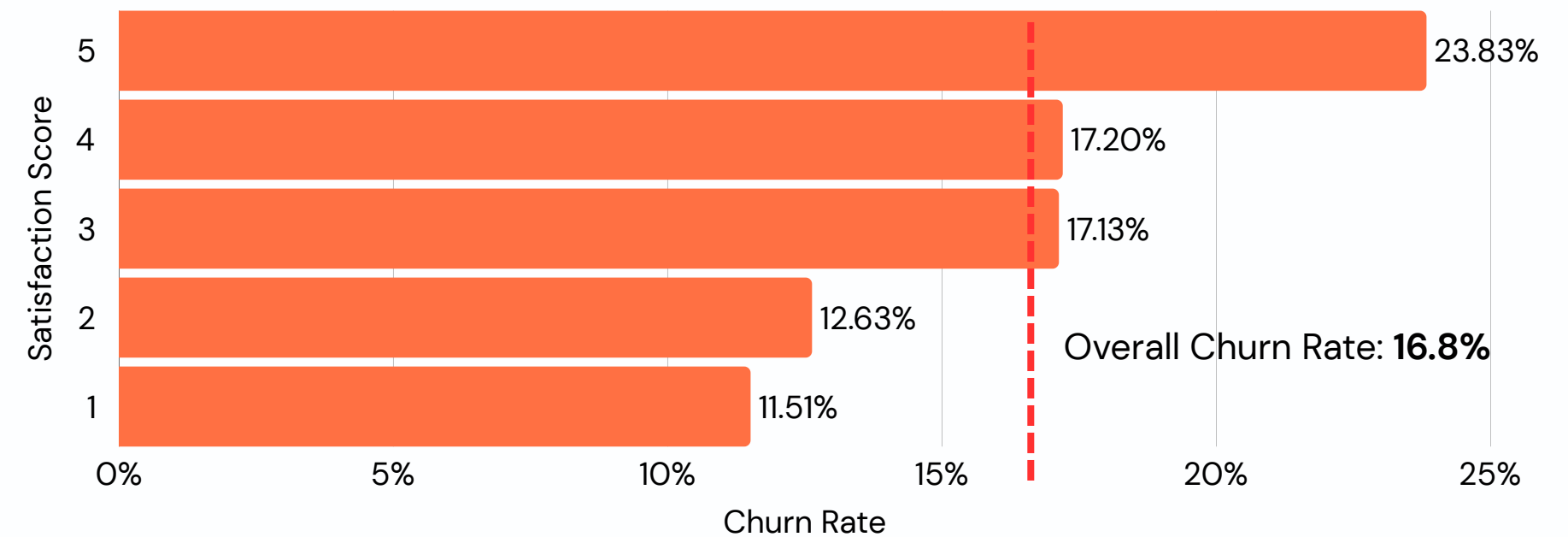
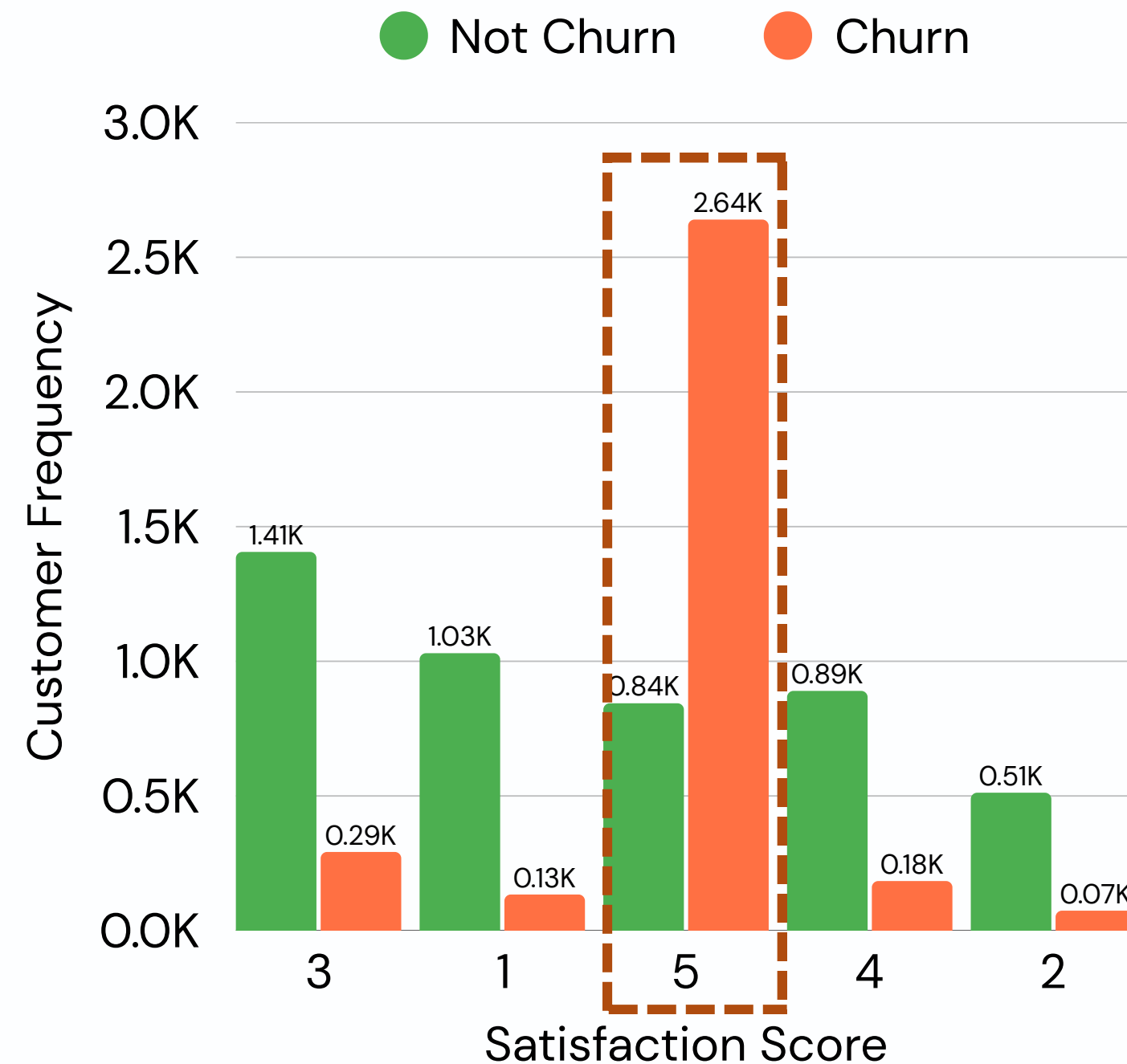
- The highest churn rate is in the Mobile Phone category at 27.4%, followed by Fashion at 19.5%
- Mobile Phone churn rate is 1.63× higher than the overall churn rate (16.8%).
- 5.6 times higher than customers in the Grocery category (lowest churn rate).

Retention strategies should focus on customers who order mobile phones and fashion items, as they are at the highest risk of churn



Exploratory Data Analysis

Churn Rate by Satisfaction Score



Insights:

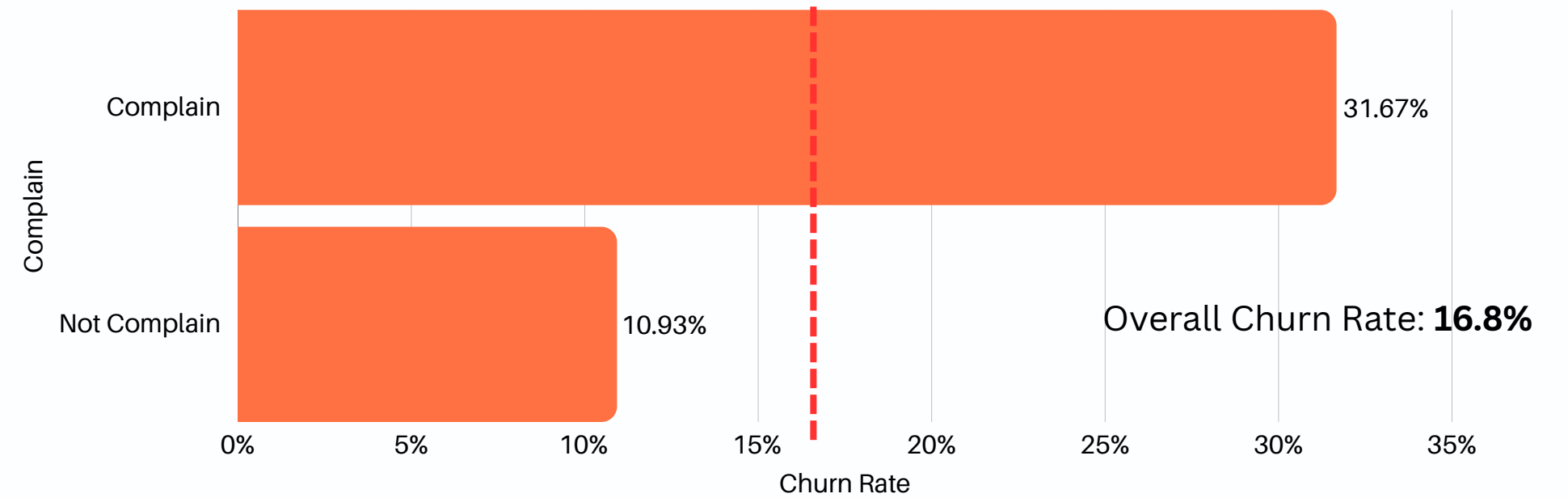
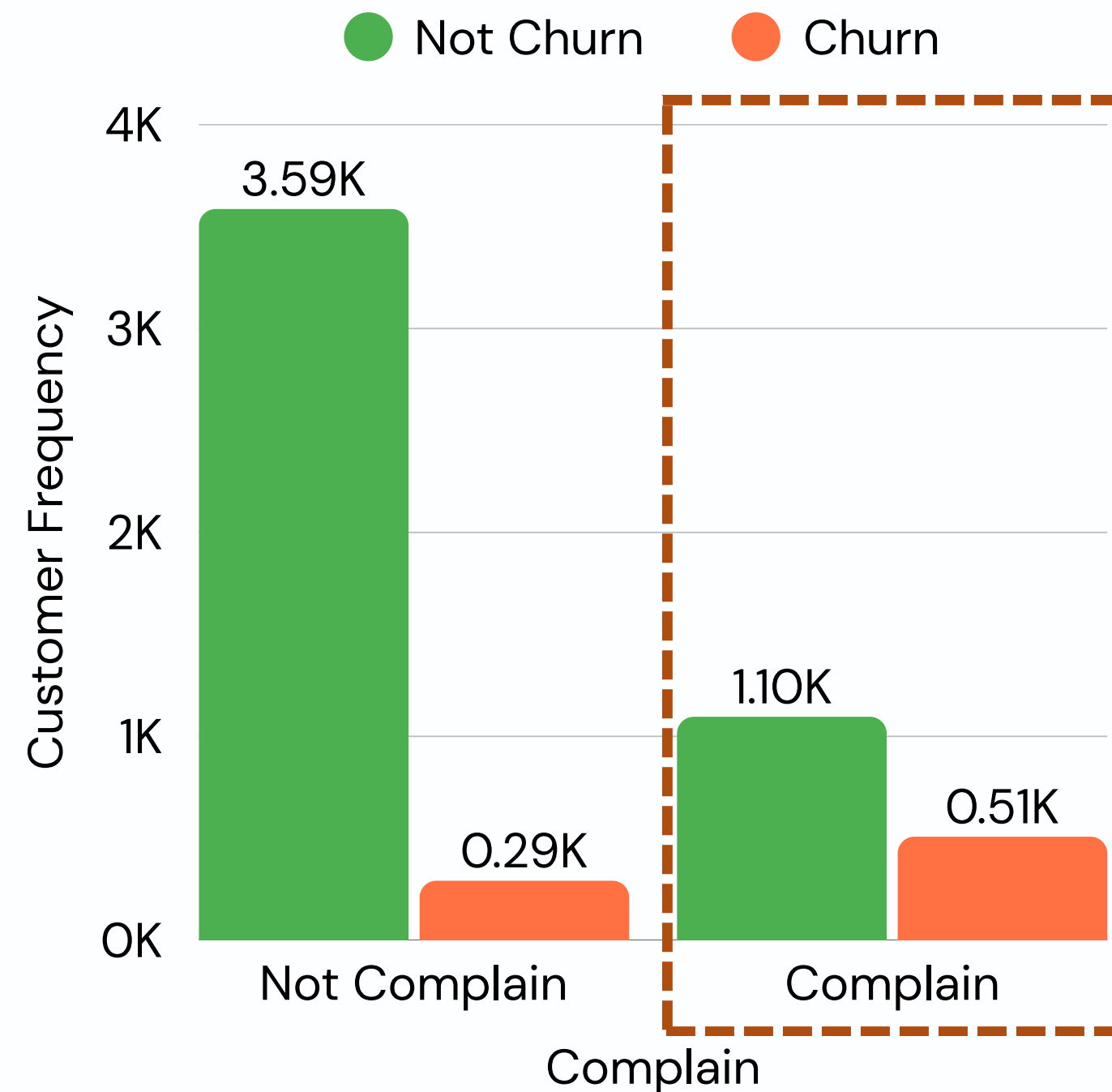
- 23.8% of customers with a satisfaction score of 5 (very good) tend to churn
- This rate is 1.42× higher than the overall churn rate (16.8%).
- 2.07 times higher than customers with a satisfaction score of 1 (lowest churn rate).

Investigate why highly satisfied customers are churning.



Exploratory Data Analysis

Churn Rate by Complain



Insights:

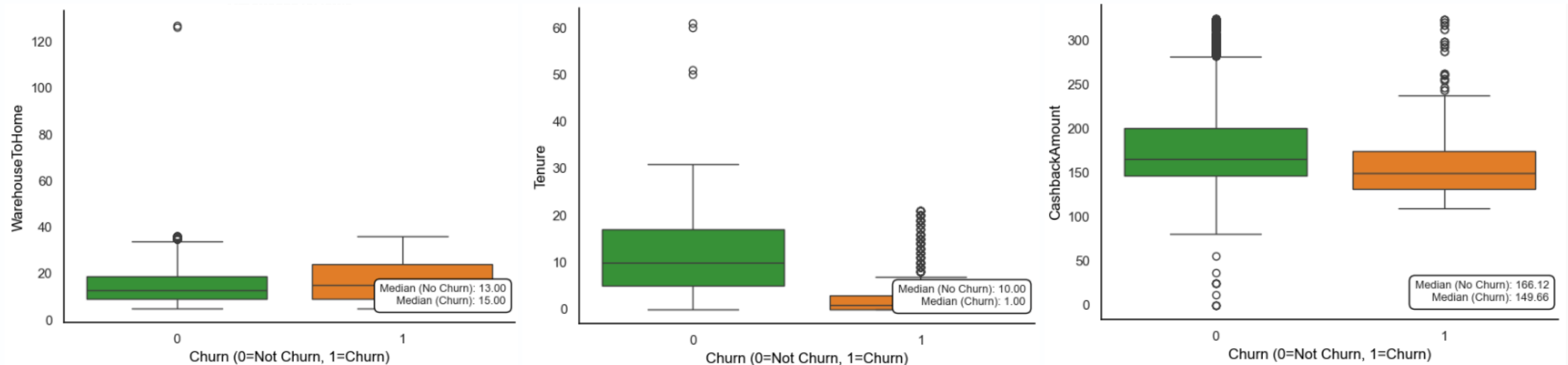
- 31.7% of customers who made a complaint churned — nearly 3× higher than those who did not complain (10.9%).
- Complainers make up a significant portion of churned customers despite being fewer in number.
- Customers who complain are at significantly higher risk of leaving.

Prioritizing strategies that improve complaint resolution speed, empathy in responses, and post-resolution engagement is essential for reducing churn among dissatisfied customers.



Exploratory Data Analysis

Warehouse To Home, Tenure, and Cashback Amount



Insights:

- From warehouse to home boxplot, churned customers are somewhat more spread out across higher distance values. This suggests that longer distances may slightly increase the risk of churn, possibly due to delivery challenges or perceived inconvenience.
- Customers who churn tend to have significantly shorter tenure compared to those who stay. The boxplot shows that the median tenure for churned customers is only 1 month, while for non-churned customers it is 10 months.
- Retained customers receive slightly more cashback, which may help reduce churn risk.



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MODELING





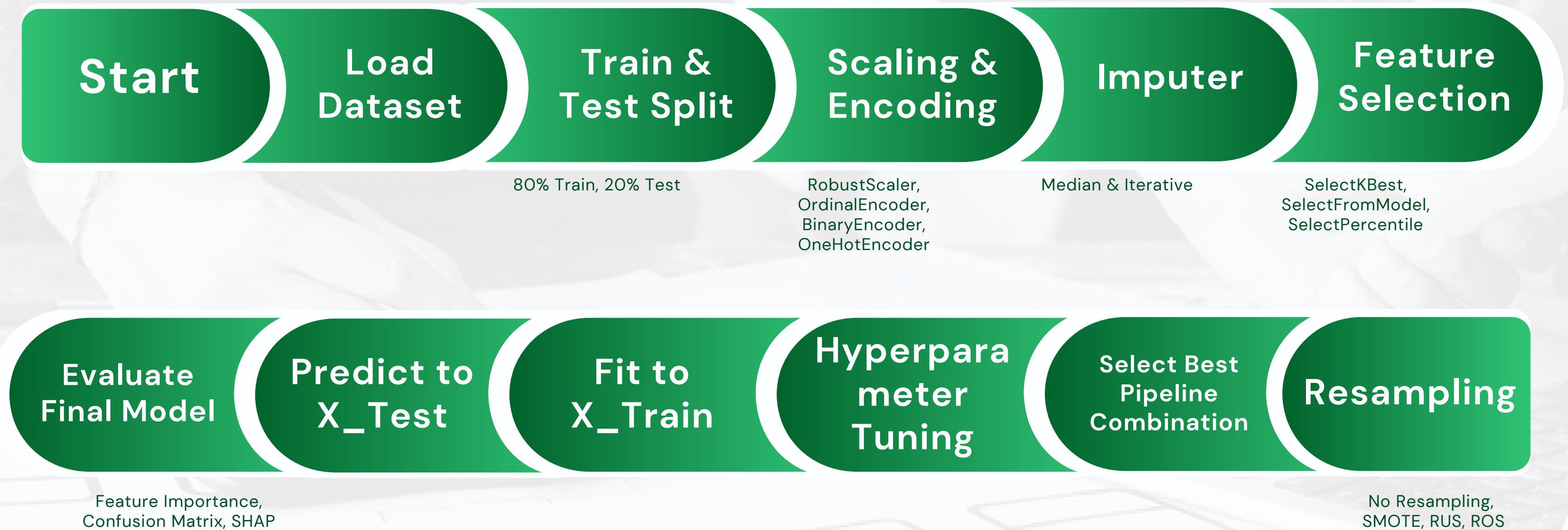
Drop Features

Column	Reason
<i>CustomerID</i>	Customer ID is removed because it is only a unique identifier with no predictive power, and keeping it may cause data leakage and lead the model to memorize patterns instead of learning meaningful features.
<i>Churn</i>	The Churn column is removed from the features because it is the target variable to be predicted, and including it during training would cause data leakage.

Modeling



End-to-End Modeling Process





Feature Engineering

Scaler

Robust Scaler was applied to all numerical features using the median, making it less sensitive to outliers and ensuring that extreme values have less influence, resulting in more stable scaling.

Encoder

OneHotEncoder was applied to the categorical features **PreferredLoginDevice** and **Gender**, transforming each category into a separate binary column.

BinaryEncoder was applied to the categorical features **PreferredPaymentMode**, **PreferredOrderCat**, **MaritalStatus**, and **CountOfAddress**, converting each category into a binary representation of integers assigned to each unique value.

OrdinalEncoder was applied to the categorical features **SatisfactionScore** and **CityTier**, mapping each category to a corresponding integer based on its inherent order.

Modeling



Benchmarking Models

Benchmarking is the process of comparing different methods or models to identify the most effective and efficient solution for the given problem. To ensure optimal model performance, we conducted a thorough benchmarking process across multiple dimensions.

Features Selection Methods	Resampling Techniques	Imputation Methods	Models
SelectKBest	No Resampling	Median Imputer	Logistic Regression
			Decision Tree
			Random Forest
SelectFromModel	SMOTE		XGBoost
			LightGBM
	Random Oversampling (ROS)	Iterative Imputer	K-Nearest Neighbors (KNN)
SelectPercentile	Random Undersampling (RUS)		Support Vector Machine (SVM)

Modeling

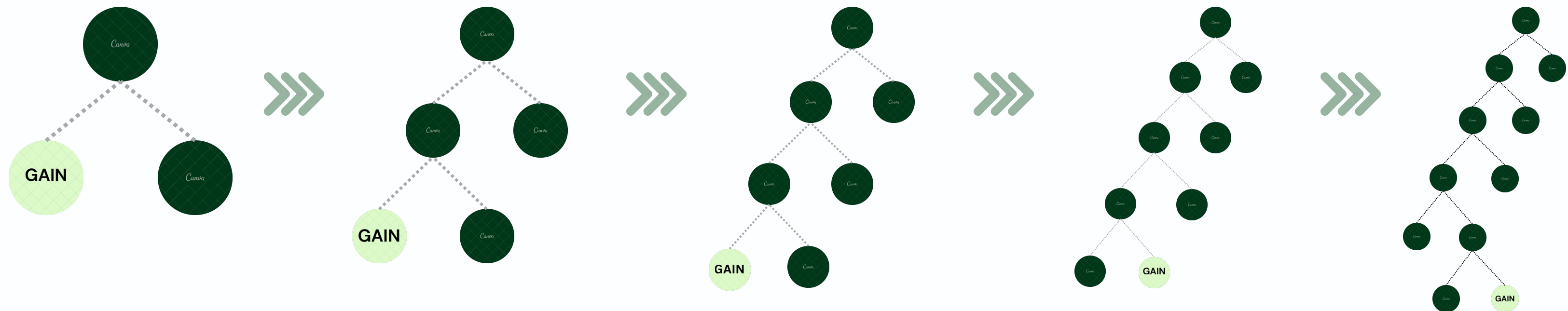


Final Model Selection to Tune

Sampler	Model	Avg F2 Score	F2 Score Std	Avg Fit Time	Avg Score Time	K Features
Random Over Sampler	LGBMClassifier	0.851406	0.010308	0.128294	0.011148	24

LGBMClassifier

LGBMClassifier is a gradient boosting framework that builds an ensemble of decision trees using a leaf-wise growth strategy to solve classification problems efficiently.



Modeling



Hyperparameter Tuning

Parameter	Information	Range
n_estimators	The number of boosting iterations (trees) the model will build. Higher values can improve performance but may increase training time and risk of overfitting.	100 - 400
max_depth	The maximum depth of each decision tree. Controls model complexity — deeper trees can capture more patterns but may also overfit.	3 - 12
learning_rate	A shrinkage parameter that scales the contribution of each tree. Lower values make learning slower but often improve generalization.	0.01 - 0.2
subsample	Controls the fraction of training samples (rows) used to grow each tree. Lower values in this range introduce randomness, helping reduce overfitting.	0.6 - 0.4
colsample_bytree	Controls the fraction of features (columns) used to grow each tree. Lower values in this range make each tree consider fewer features, which also reduces overfitting.	0.6 - 0.4

F2 Score (Before Hyperparameter Tuning)	F2 Score (After Hyperparameter Tuning)
85%	91%

F2 Score Improvement Through Hyperparameter Tuning



Confusion Matrix

Before Hyperparameter Tuning

Actual	Not Churn (0)	Churn (1)
	Not Churn (0)	Churn (1)
Not Churn (0)	897	39
Churn (1)	19	171

Actual Not Churn (0):

- **Correctly Predicted** as Not Churn (TN): **897**
- **Wrongly Predicted** as Churn (FP): **39**

Actual Churn (1):

- **Correctly Predicted** as Churn (TP): **171**
- **Wrongly Predicted** as Not Churn (FN): **19**



Confusion Matrix

After Hyperparameter Tuning

Actual \ Predicted	Not Churn (0)	Churn (1)
Not Churn (0)	921	15
Churn (1)	17	173

Actual Not Churn (0):

- **Correctly Predicted** as Not Churn (TN): 921
- **Wrongly Predicted** as Churn (FP): 15

Actual Churn (1):

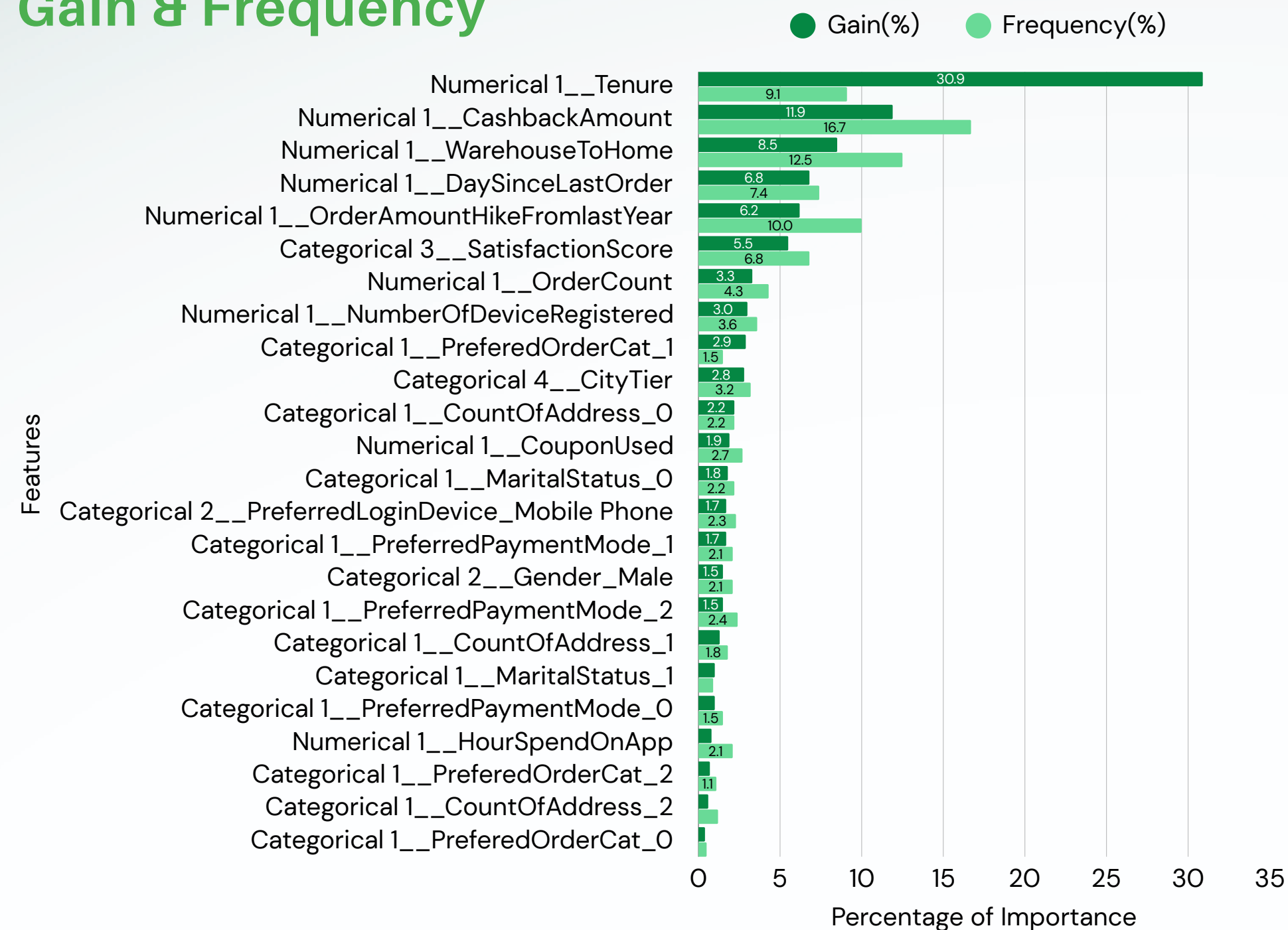
- **Correctly Predicted** as Churn (TP): 173
- **Wrongly Predicted** as Not Churn (FN): 17

*In this project, our focus is minimizing false negatives. Before tuning, the model had 19 false negatives, and after tuning, **it was reduced to 17.***



Feature Importance

Gain & Frequency



Insights :

- **Tenure → High Gain, Low Frequency**

The length of time a customer has been with the company is a strong churn predictor, even though it's not used in many splits.

- **Cashback Amount → High Gain, High Frequency**

Cashback patterns are both frequently used and impactful.

- **Warehouse to Home Distance → High Gain, Moderate Frequency**

Delivery distance matters. Customers who live farther from the warehouse may experience slower delivery times, potentially increasing dissatisfaction and churn risk.

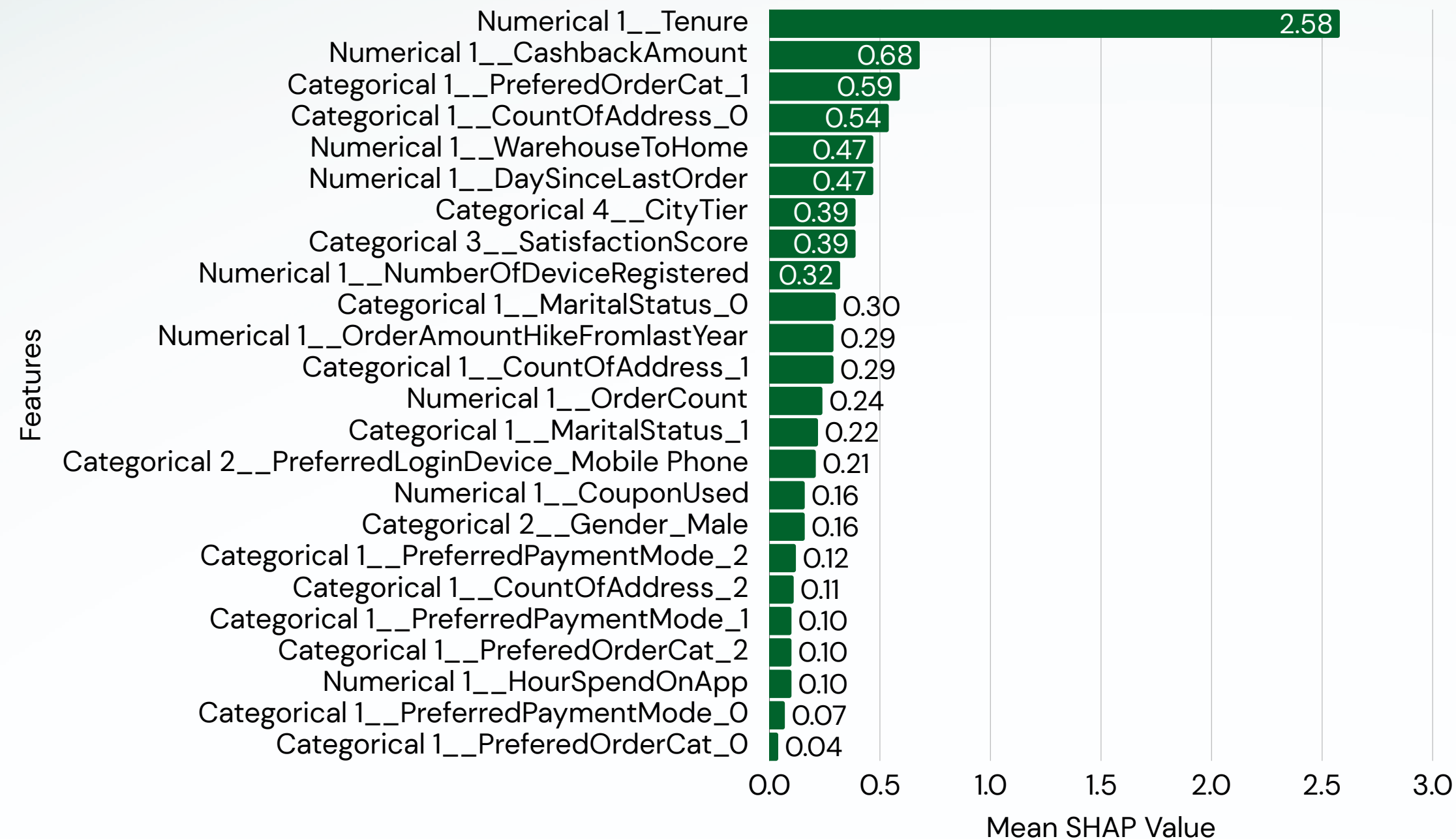
- **Day Since Last Order → High Gain, Moderate Frequency**

The recency of a customer's last order is a strong behavioral indicator. Longer gaps since the last order often signal declining engagement and an increased chance of churn.

A high gain but low frequency feature means **rarely used, but game-changing when it appears**. Meanwhile a high frequency but low gain feature means **often used, but each split has modest impact**.

SHAP Summary

for Customer Churn Prediction



Insights :

- **Tenure**

The most influential feature. Even small changes in tenure can significantly affect churn probability. Lower tenure typically indicates a higher likelihood of churn.

- **CashbackAmount**

Lower cashback amounts are generally associated with increased churn risk.

- **PreferedOrderCat_1**

Certain product categories show a clear correlation with churn behavior.

- **WarehouseToHome**

Delivery distance can affect customer satisfaction and churn rates.

- **DaySinceLastOrder**

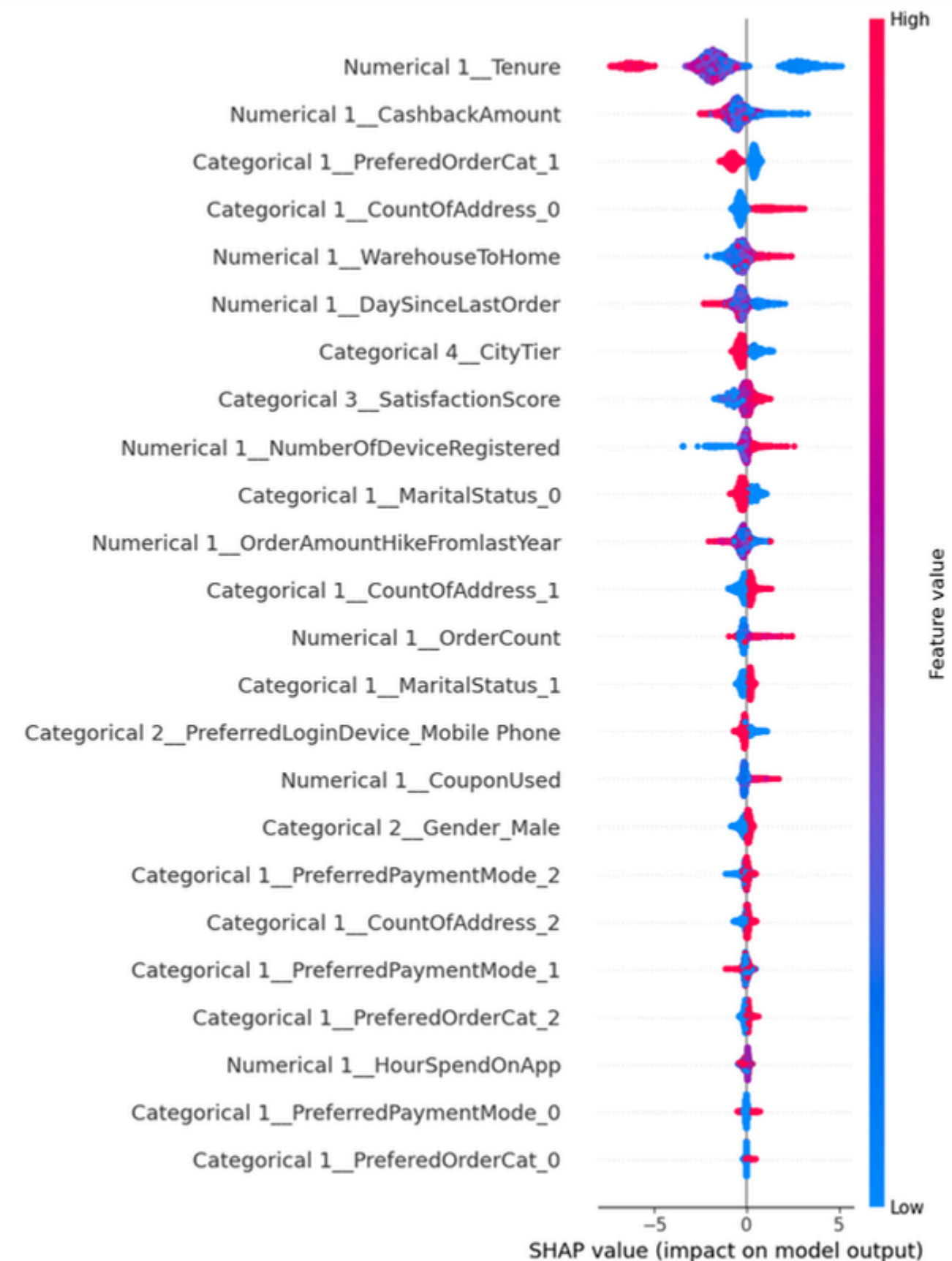
Purchase recency is a strong indicator of customer engagement.

SHAP Summary

for Customer Churn Prediction

Insights :

- Three most influential features in predicting customer churn are **Numerical_1_Tenure**, **Numerical_1_CashbackAmount**, and **Categorical_1_PreferedOrderCat_1**.
- Customers with **lower tenure (blue)** have positive SHAP values, indicating a **higher likelihood of churn**.
- For CashbackAmount, **lower cashback (blue)** tends to have positive SHAP, meaning less cashback is **associated with higher churn**.
- Certain order categories appear to be **linked with a higher churn risk**, as indicated by their positive SHAP values.



Modeling



Model Constraint - Numerical Features

Feature Name	Data Type	Min	Max
Tenure	Integer	0	61
Warehouse To Home	Integer	5	126
Hour Spend On App	Float	0.0	5.0
Number Of Device Registered	Integer	1	6
Order Amount Hike From LastYear	Integer	11	26
Coupon Used	Integer	0	16
Order Count	Integer	1	16
Day Since Last Order	Integer	0	46
Cashback Amount	Float	0.00	324.99

Modeling



Model Constraint - Categorical Features

Feature Name	Categories
Preferred Login Device	Mobile Phone, Computer
City Tier	1, 2, 3
Preferred Payment Mode	Debit Card, Credit Card, E wallet, Cash on Delivery, UPI
Gender	Female, Male
Prefered Order Cat	Mobile Phone, Laptop & Accessory, Grocery, Fashion, Others
Satisfaction Score	1, 2, 3, 4, 5
Marital Status	Single, Married, Divorced
Complain	0, 1 (0 = not complain, 1 = complain)
Count Of Address	1-2, 3, 4-6, 7+

Customer *Churn* Prediction



Please input customer features

Tenure (in months)
12

Warehouse To Home (in km)
20

Hour Spent on App (per day)
1.00

Number of Devices Registered
2

Order Amount Hike From Last Year (%)
15

Coupons Used
3

Order Count
5

Days Since Last Order
10

Cashback Amount
50.00

Customer Churn Predictor

This web predicts the likelihood of a customer churning

Customer's Features

	0
Tenure	12
PreferredLoginDevice	Mobile Phone
CityTier	1
WarehouseToHome	20
PreferredPaymentMode	Debit Card
Gender	Female
HourSpendOnApp	1.0
NumberOfDeviceRegistered	2
PreferredOrderCat	Mobile Phone
SatisfactionScore	1

Prediction Result

Churn: **No** – This customer is likely to stay.

Probability of Churn: 0.00

As the output of our model development, we have deployed the customer churn prediction model on a website.

[Access the website here](#)

< Manage app



Model *Implementation*



Model Training

The model is trained using historical customer data

Observation Window

Retention Opportunity Window

NOW



Churn Prediction

It identifies customers most likely to churn within a specific time frame (e.g., next month).



Early Warning

These predictions alert the business in advance about high-risk customers.



Retention Actions

The customer marketing team can use this time-window to offer targeted promotions, discounts, better services, or personalized follow-ups to retain customers.



Business Evaluation

Cost Assumptions:

False Positive

\$

False Negative

\$7

Without Machine Learning

Actual	Churn	936	
	Not Churn	190	
		Not Churn	Churn
		Predicted	

$$190 \times \$7 = \$1330$$

With Machine Learning

Actual	Churn	921	15
	Not Churn	17	173
		Not Churn	Churn
		Predicted	

$$(17 \times \$7) + (15 \times \$1) = \$134$$

By leveraging the model, cost are reduced by nearly tenfold compared to scenarios without a model, significantly improving cost efficiency and enhancing strategic decision-making for customer management.



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CONCLUSION AND RECOMMENDATION





Conclusion

Project & Methodology

This project developed a robust and interpretable ML pipeline for predicting e-commerce churn, combining LightGBM, Median Imputer, SelectKBest (k=24), and RandomOverSampler. The model achieved a 91% F2 Score, focusing on minimizing false negatives to capture high-risk customers

Key Findings

Churn is driven by low cashback, short tenure, and recent complaints, with an unexpected increase among highly satisfied customers (score 5). Higher churn in lower-tier cities highlights geographic and service-level gaps, suggesting the need for targeted retention strategies.

Business Recommendation



Boost Cashback Incentives

Offer targeted and tiered cashback programs, especially for new or high-risk customers, to encourage early engagement and reward loyalty.



Improve Delivery Experience

Address higher churn in remote areas by offering free/discounted shipping and optimizing logistics for customers far from warehouses.



Investigate Satisfaction Score Anomalies

Analyze why high-satisfaction customers still churn by look into data quality, possible rating manipulation, and consider adopting a more robust feedback system.



Business Recommendation



Re-Engage Low-Order Customers

Identify customers with low order frequency and target them with reminders, personalized offers, or “win-back” campaigns to drive repeat purchases.



Act Quickly on Complaints

Enhance complaint handling by resolving issues faster, analyzing complaint types, and rewarding feedback to restore customer trust and reduce churn risk



Model Recommendation

Adding Complaint Type as a Churn Predictor

Customers who file complaints are more likely to churn. Including a “complaint type” feature such as late delivery, damaged items, or refund delays can help the model better predict churn and give the business clearer insights on which issues to fix first.

Future Exploration With Alternative Models

Given more time and access to a higher-specification device, it is suggested to explore alternative algorithms such as XGBoost and CatBoost to compare their performance with LightGBM. While LightGBM was chosen for its efficiency and competitive accuracy, CatBoost often require longer training times and greater computational resources.





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**THANK YOU FOR YOUR
ATTENTION!**